

STATES OF MATTER & CHANGES OF STATE

KEY VOCABULARY

Melting: solid → liquid as particles gain energy.

Boiling/evaporating: liquid → gas.

Condensing: gas → liquid.

State symbols: (s) solid, (l) liquid, (g) gas, (aq) dissolved.

COMMON MISCONCEPTIONS

~~— Particles expand when heated.~~

✓ Particles gain energy and move more — they don't grow.

~~— Mass changes during a state change.~~

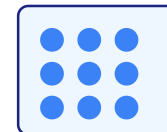
✓ Mass is conserved; only arrangement/energy changes.

~~— Gases have no mass.~~

✓ Gas particles still have mass.

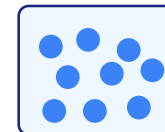
PARTICLE ARRANGEMENT

SOLID



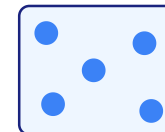
fixed, regular, touching

LIQUID



close, random, can flow

GAS



far apart, fast, random

1 ARRANGEMENT

Describe the arrangement of particles in a **solid**. [2 marks]

2 NAME THE CHANGE

Name the change of state from **gas to liquid**. [1 mark]

3 ENERGY & MELTING

Explain, using particles, what happens when a solid melts. [3 marks]

4 STATE SYMBOLS

Give the state symbol for a substance **dissolved in water**. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

When a substance is heated, particles gain ____ so the forces between them become ____ enough to break.

energy

closer

weaker

stronger

6 LIMITATIONS

Suggest one limitation of the simple particle model. [1 mark]
